4. POLARNET - Decentralized Arctic Climate Data Marketplace

To address the limited coverage and government-controlled access of traditional systems, this mission proposes a constellation of IoT relay satellites for the Arctic region. These satellites collect vital climate data like ice thickness and methane emissions from remote sensors to populate a decentralized marketplace where researchers can purchase verified data directly through smart contracts.

Satellite Requirements

- **Orbit:** Molniya orbit (63.4° inclination, 12-hour period) OR polar LEO constellation
- **Coverage:** 24/7 coverage above 70°N latitude
- **Payload:** LoRa/UHF receiver for IoT sensors, store-and-forward data relay
- **Power:** 30W average (extreme thermal environment: -40°C to +60°C)
- **Mass:** <25 kg per satellite (6U CubeSat)
- **Comms:** UHF uplink from sensors, S-band downlink to ground stations

Web3 System

Data Flow: Arctic sensors (ice buoys, weather stations, wildlife trackers) → Satellite collects data during pass → Uploads to IPFS with cryptographic hash → Smart contract verifies data integrity → Researchers purchase access via tokens → Sensor operators earn revenue → Data becomes public after 6 months

Key Smart Contracts:

1. **DataRegistry** - Catalogs all sensor data with IPFS hashes, timestamps, GPS coordinates
2. **DataMarketplace** - Handles purchases (researchers pay tokens for specific datasets)
3. **SensorVerification** - Uses multiple satellite confirmations to verify data authenticity
4. **RevenueDistribution** - Splits payments between sensor operators and satellite operators

Round 1 constraints:

1. Mission Objectives & Success Criteria

Every design must start with the **"Why."** In a hackathon, your mission objective isn't just a goal; it's a boundary.

- **Operational Lifetime:** How long must the satellite survive? (e.g., 6 months for a tech demo vs. 5 years for a weather satellite).
- **Target Orbit:** This dictates almost everything else.
    - **LEO (Low Earth Orbit):** Lower latency, higher resolution, but high atmospheric drag.
    - **GEO (Geostationary Orbit):** Constant coverage of one area, but extremely high launch costs and radiation.
- **Data Latency:** Does the data need to be "real-time" (for disaster relief) or can it be stored and forwarded once a day?

2. Payload Constraints

The payload is the "customer" of the satellite; everything else (power, thermal, comms) exists just to keep it happy.

- **SWaP-C:** This is the golden rule of satellite design: **S**ize, **W**eight, **a**nd **P**ower - **C**ost.
- **Mass & Volume:** Most hackathons use the "U" standard (1U = 10x10x10 cm). You'll likely be restricted to a 3U or 6U form factor.
- **Duty Cycle:** Can the payload run 24/7, or does it need to shut down to recharge batteries while the satellite is in the Earth's shadow?

3. Budget & Resource Estimation

In a real-world scenario (and a good hackathon), "unlimited money" isn't a thing.

- **Total Cost of Ownership (TCO):** You must account for:
    1. **Development:** Components and labor.
    2. **Launch:** Often calculated per kilogram (e.g., $50,000/kg for LEO).
    3. **Operations:** Ground station fees to talk to your satellite.
- **COTS vs. Custom:** Using **Commercial Off-The-Shelf (COTS)** parts is cheaper and faster but might not be "space-hardened" against radiation compared to custom, radiation-hardened components.

4. Technical Constraints & Requirements

1. The Arctic Access Problem

Traditional satellites often struggle with Arctic coverage because most orbit around the equator.

- **The Molniya Solution:** We use a **Molniya Orbit**—a highly "egg-shaped" path that lets the satellite "dwell" over the North Pole for most of its 12-hour orbit. This provides the 24/7 coverage needed to track fast-moving ice changes and methane leaks.
- **Store-and-Forward:** Arctic sensors (buoys, trackers) are low-power. The satellite acts like a **"Mailman in Space"**: it picks up small data packets as it passes over the sensors and "carries" them until it reaches a ground station to drop them off.

2. IoT & The Decentralized Marketplace

- **The Sensors:** We deploy a "Marine/Land IoT" network. These are small, rugged, solar-powered buoys that measure **ice thickness** and **methane emissions**.
- **Direct-to-Satellite:** Using **LoRa/UHF** technology, these sensors can talk directly to our CubeSats from thousands of kilometers away without needing a local internet connection.
- **The Marketplace:** Instead of the data being hidden in a government database, it is uploaded to a **Blockchain Marketplace**.
    - Researchers buy access using tokens.
    - The money goes directly to the people who maintain the sensors (indigenous communities, local scientists).

3. The Trust Layer (Smart Contracts)

- **Data Verification:** How do we know a sensor isn't lying? The **SensorVerification** contract compares data from multiple satellites. If one buoy reports "ice melting" but three others nearby say "ice stable," the system flags the anomaly.
- **Public Commons:** To prevent "data hoarding," all climate data becomes **Public Domain** after 6 months. This ensures that even researchers without funding eventually get the data they need to save the planet.

4. Technical & Operational Readiness

- **Small but Mighty:** We use **6U CubeSats** (<25 kg). These are cheap to launch but large enough to house the powerful UHF receivers and S-band transmitters needed for high-speed data delivery.
- **Extreme Thermal Protection:** Arctic satellites face intense temperature swings (-40°C to $+60\text{°C}$). The mission uses specialized **Passive Thermal Control** (heat blankets and radiators) to survive the brutal polar environment.